# Lab 3 — Full Explanation: Parameter Estimation and Unbiasedness of Estimators

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This file explains every formula in the notebook. Each one is derived from first principles, every step is justified, and the simulation results are explained by the theory. It is meant as preparation for the defence, where every team member must know every derivation.

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# 0. Theory toolbox

## 0.1 Sample, statistic, estimator

A **sample** $X_1, ..., X_n$ is a set of i.i.d. (independent, identically distributed) random variables with a common distribution that depends on an unknown parameter $\theta$.

A **statistic** is any function of the sample that does not involve unknown parameters, e.g. the sample mean

$$\overline{\mathbf{X}} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

or the sample standard deviation

$$S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{\mathbf{X}})^2}.$$

An **estimator** $\hat{\theta}$ of $\theta$ is a statistic used to guess $\theta$. It is random, because it depends on the random sample.

- **Unbiased:** $\mathsf{E}\hat{\theta} = \theta$ for every value of $\theta$. On average, the estimator is right.
- **Bias:** $\text{Bias}(\hat{\theta}) = \mathsf{E}\hat{\theta} - \theta$.
- **Asymptotically unbiased:** $\text{Bias}(\hat{\theta}) \to 0$ as $n \to \infty$.
- **Consistent:** $\hat{\theta} \to \theta$ in probability as $n \to \infty$.
- **Mean squared error:** $\text{MSE}(\hat{\theta}) = \mathsf{E}(\hat{\theta} - \theta)^2 = \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$.

## 0.2 Two facts about the sample mean (used everywhere)

If $\mathsf{E}X_i = \mu$ and $\text{Var}\, X_i = \sigma^2$:

$$\mathsf{E}\overline{\mathbf{X}} = \frac{1}{n}\sum_{i=1}^{n}\mathsf{E}X_i = \frac{1}{n}\cdot n\mu = \mu \qquad \text{(linearity of expectation)}$$

$$\text{Var}\,\overline{\mathbf{X}} = \frac{1}{n^2}\,\text{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}\, X_i = \frac{\sigma^2}{n}$$

In the second line, the variance of a sum equals the sum of the variances **because of independence**: all covariances are 0. The factor $1/n^2$ appears because $\text{Var}(cY) = c^2\,\text{Var}\, Y$.

So $\overline{\mathbf{X}}$ is an unbiased estimator of $\mu$, and its standard deviation is $\sigma/\sqrt{n}$. This $1/\sqrt{n}$ is why **all interval lengths shrink like $1/\sqrt{n}$**.

## 0.3 Confidence interval: definition and meaning

A **confidence interval of level $1-\alpha$** is a pair of statistics $L(\mathbf{X}) \le U(\mathbf{X})$ such that

$$\mathsf{P}_\theta(L(\mathbf{X}) \le \theta \le U(\mathbf{X})) = 1 - \alpha \quad \text{for every} \quad \theta.$$

The endpoints are random and $\theta$ is fixed. "95% confidence" means: if we repeat the experiment many times, about 95% of the intervals we build will contain the true $\theta$. It does **not** mean "$\theta$ is in this particular interval with probability 0.95". After the data are observed, $\theta$ either is or is not in it.

This is exactly what the lab checks by simulation. We generate $m$ samples, build $m$ intervals, and count the fraction that contain $\theta$. That fraction is the **empirical confidence** (also called **coverage**).

When the probability above only approximately equals $1-\alpha$ (for example, via the CLT), the interval is called **approximate** or **asymptotic**. Its real coverage can differ from $1-\alpha$, and measuring that difference is the whole point of Part I.

## 0.4 Quantiles, and why $\beta = 1 - \alpha/2$

The ***p*-quantile** $q_p$ of a continuous distribution with CDF $F$ is the number with $F(q_p) = p$. Notation used in the notebook:

- $z_\beta$: quantile of $\mathcal{N}(0,1)$, so $\Phi(z_\beta) = \beta$.
- $t_{n-1,\beta}$: quantile of the Student distribution with $n-1$ degrees of freedom.
- $\chi^2_{k,p}$: quantile of $\chi^2_k$.

For $Z \sim \mathcal{N}(0,1)$, the density is symmetric, so $\Phi(-z) = 1 - \Phi(z)$. Then

$$\mathsf{P}(|Z| \le z_\beta) = \Phi(z_\beta) - \Phi(-z_\beta) = \Phi(z_\beta) - (1 - \Phi(z_\beta)) = 2\Phi(z_\beta) - 1 = 2\beta - 1.$$

We want this to equal $1-\alpha$: $2\beta - 1 = 1 - \alpha \iff \beta = 1 - \alpha/2$. Equivalently, we cut off probability $\alpha/2$ in **each** tail. The values used in the lab:

| $\alpha$ | $1-\alpha$ | $\beta$ | $z_\beta$ |
|---|---|---|---|
| 0.10 | 0.90 | 0.95 | 1.645 |
| 0.05 | 0.95 | 0.975 | 1.960 |
| 0.01 | 0.99 | 0.995 | 2.576 |

## 0.5 The pivot method (how every interval is built)

A **pivot** is a function $Q(\mathbf{X}, \theta)$ of the data **and** the parameter whose distribution is known and **does not depend on $\theta$**. The recipe is:

1. Find a pivot $Q$, exactly or approximately.
2. Take quantiles $a, b$ of its distribution with $\mathsf{P}(a \le Q \le b) = 1 - \alpha$.
3. Solve the inequality $a \le Q(\mathbf{X}, \theta) \le b$ for $\theta$. This gives $L \le \theta \le U$.

Step 3 is an **equivalent transformation** of events, so the probability stays $1-\alpha$. The four methods of Problem 1 are four choices of pivot:

| Method | Pivot | Distribution |
|---|---|---|
| 1 | $2n\overline{\mathbf{X}}/\theta$ | exactly $\chi^2_{2n}$ |
| 2 | $\sqrt{n}(\overline{\mathbf{X}} - \theta)/\theta$ | $\approx \mathcal{N}(0,1)$, but step 3 is not done ($\theta$ is left in the endpoints) |
| 3 | same as 2 | $\approx \mathcal{N}(0,1)$, step 3 done properly |
| 4 | $\sqrt{n}(\overline{\mathbf{X}} - \theta)/S$ | $\approx t_{n-1}$ |

## 0.6 Distributions used

**Exponential $\mathscr{E}(\lambda)$.** Density $f(x) = \lambda e^{-\lambda x}$ for $x \ge 0$; CDF $F(x) = 1 - e^{-\lambda x}$.

**Poisson $\mathscr{P}(\theta)$.** $\mathsf{P}(X = k) = e^{-\theta}\theta^k/k!$ for $k = 0, 1, 2, \ldots$

**Gamma $\Gamma(k, 1)$.** Density $\frac{x^{k-1}e^{-x}}{\Gamma(k)}$ for $x \ge 0$. It is the distribution of a sum of $k$ independent $\mathscr{E}(1)$ variables.

**Chi-square $\chi^2_k$.** The distribution of $Z_1^2 + \ldots + Z_k^2$ with $Z_i$ i.i.d. $\mathcal{N}(0,1)$. Its density is

$$f_k(y) = \frac{y^{k/2-1}e^{-y/2}}{2^{k/2}\Gamma(k/2)}, \quad y \geq 0,$$

which is a Gamma distribution with shape $k/2$ and scale 2. Its MGF is

$$M(t) = (1-2t)^{-k/2}, \quad t < \frac{1}{2}.$$

For $k = 2$: $f_2(y) = \frac{1}{2}e^{-y/2}$, i.e. $\chi^2_2$ **is** the exponential distribution with mean 2.

**Student $t_k$.** If $Z \sim \mathcal{N}(0,1)$ and $V \sim \chi^2_k$ are **independent**, then $Z/\sqrt{V/k} \sim t_k$. It is symmetric and has heavier tails than $\mathcal{N}(0,1)$, so $t_{k,\beta} > z_\beta$, and $t_{k,\beta} \to z_\beta$ as $k \to \infty$. At $\beta = 0.975$: $t_4 = 2.776$, $t_9 = 2.262$, $t_{29} = 2.045$, $t_{99} = 1.984$, compared with $z = 1.960$.

**Moment generating function (MGF).** $M_X(t) = \mathsf{E}e^{tX}$. Two properties are used:

- **Uniqueness:** if two random variables have the same MGF near 0, they have the same distribution.
- **Independence:** if $X, Y$ are independent, then $M_{X+Y}(t) = M_X(t)M_Y(t)$, because $\mathsf{E}e^{t(X+Y)} = \mathsf{E}e^{tX}\,\mathsf{E}e^{tY}$.

## 0.7 Limit theorems

- **Law of Large Numbers (LLN):** $\overline{\mathbf{X}} \to \mu$. Also $S^2 \to \sigma^2$, so $S \to \sigma$ (in probability).
- **Central Limit Theorem (CLT):** if $\sigma^2 < \infty$, then

  $$\frac{\sqrt{n}(\overline{\mathbf{X}} - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0,1).$$

  For finite $n$ this is only an approximation. The approximation is worse when the distribution is **skewed** or **discrete**, which is exactly the case for the exponential and Poisson distributions.
- **Slutsky's theorem:** if $Y_n \xrightarrow{d} Y$ and $c_n \to c$ in probability, then $Y_n/c_n \xrightarrow{d} Y/c$. It lets us replace $\sigma$ by $S$ in the CLT, which justifies method 4 for large $n$.

## 0.8 Monte Carlo estimation of the coverage (the role of $m$)

For one method, let $I_j = 1$ if the $j$-th interval contains $\theta$, otherwise 0. The $I_j$ are i.i.d. Bernoulli($p$), where $p$ is the **true** coverage probability. The empirical confidence is

$$\hat{p} = \frac{1}{m}\sum_{j=1}^{m} I_j, \qquad \mathsf{E}\hat{p} = p, \qquad \operatorname{Var}\hat{p} = \frac{p(1-p)}{m}.$$

(This is the formula $\operatorname{Var}\overline{\mathbf{X}} = \sigma^2/n$ from §0.2, with Bernoulli variance $p(1-p)$.) Therefore:

- ***m* does not change *p*.** It only controls how precisely we estimate it. The Monte Carlo standard error is $\sqrt{p(1-p)/m}$.
- ***n* changes *p* itself**, because it changes how good the approximation is. For an exact method, $p = 1 - \alpha$ for every $n$.

For $p = 0.95$:

| $m$ | $10^2$ | $10^3$ | $10^4$ | $10^5$ |
|---|---|---|---|---|
| SE $= \sqrt{0.95 \cdot 0.05/m}$ | 0.022 | 0.007 | 0.002 | 0.0007 |

So with $m = 10^4$ (used in the lab), differences in coverage larger than about $2 \cdot 0.002 = 0.004$ are real and not noise.

# 1. Problem 1 — Exponential distribution

## 1.1 Setup and moments

$X_1, ..., X_n \sim \mathscr{E}(\lambda)$ i.i.d., parameter $\theta = 1/\lambda$. In the code, `np.random.exponential(scale=theta)` uses exactly this parametrisation: scale = mean = $\theta$. The lab uses $\theta = \texttt{ID}\ /10 = 1.1$.

**Derivation of $\mathsf{E}X = \theta$ and $\operatorname{Var} X = \theta^2$.** Integrate by parts:

$$\mathsf{E}X = \int_0^\infty x\lambda e^{-\lambda x}dx = \left[-xe^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x}dx = 0 + \frac{1}{\lambda} = \theta,$$

$$\mathsf{E}X^2 = \int_0^\infty x^2\lambda e^{-\lambda x}dx = \left[-x^2e^{-\lambda x}\right]_0^\infty + 2\int_0^\infty xe^{-\lambda x}dx = \frac{2}{\lambda}\cdot\frac{1}{\lambda} = \frac{2}{\lambda^2},$$

$$\operatorname{Var} X = \mathsf{E}X^2 - (\mathsf{E}X)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} = \theta^2.$$

So the population variance is $s^2 = \theta^2$ and the standard deviation is $s = \theta$. The mean and the standard deviation are **the same unknown number**. This is the key feature of this problem.

By §0.2, $\overline{\mathbf{X}}$ is unbiased for $\theta$ and $\operatorname{Var}\overline{\mathbf{X}} = \theta^2/n$.

**Scale property (why the coverage does not depend on $\theta$).** Write $X_i = \theta E_i$ with $E_i \sim \mathscr{E}(1)$. Then $\overline{\mathbf{X}} = \theta\,\overline{\mathbf{E}}$ and $S_X = \theta\, S_E$. Each interval below has the form "$\theta \in$ [statistic $\times$ constant, statistic $\times$ constant]". Dividing everything by $\theta$ turns it into an event about the $E_i$ only. So the coverage of all four methods is the same for every $\theta$, and the lengths are proportional to $\theta$. Changing $\theta$ would add nothing to the experiment, so it is enough to use $\theta = 1.1$.

## 1.2 Method 1 — exact pivot $2\lambda n\overline{\mathbf{X}} \sim \chi^2_{2n}$

**Step 1: $Y = 2\lambda X_i \sim \chi^2_2$.** For $y \geq 0$, use the CDF method:

$$F_Y(y) = \mathsf{P}(2\lambda X_i \leq y) = \mathsf{P}\left(X_i \leq \frac{y}{2\lambda}\right) = 1 - e^{-\lambda\cdot\frac{y}{2\lambda}} = 1 - e^{-y/2}.$$

Differentiating gives $f_Y(y) = \frac{1}{2}e^{-y/2}$, which is the $\chi^2_k$ density at $k = 2$: $\frac{y^0 e^{-y/2}}{2^1\Gamma(1)} = \frac{1}{2}e^{-y/2}$. ✓

(Alternatively, with MGFs: $M_X(t) = \int_0^\infty e^{tx}\lambda e^{-\lambda x}dx = \frac{\lambda}{\lambda - t}$ for $t < \lambda$, so $M_Y(t) = M_X(2\lambda t) = \frac{\lambda}{\lambda - 2\lambda t} = (1-2t)^{-1} = (1-2t)^{-2/2}$.)

**Step 2: a sum of independent $\chi^2_2$ variables.** We have $2\lambda n\overline{\mathbf{X}} = \sum_{i=1}^n 2\lambda X_i$, a sum of $n$ independent $\chi^2_2$ variables. By the independence property of MGFs:

$$M_{2\lambda n\overline{\mathbf{X}}}(t) = \prod_{i=1}^n (1-2t)^{-1} = (1-2t)^{-n} = (1-2t)^{-2n/2}.$$

This is the MGF of $\chi^2_{2n}$, so by uniqueness:

$$\boxed{2\lambda n\overline{\mathbf{X}} = \frac{2n\overline{\mathbf{X}}}{\theta} \sim \chi^2_{2n}}\,.$$

Intuitively, each $\chi^2_2$ is a sum of 2 squared normals, so $n$ of them give $2n$ squared normals.

This is a **pivot**: it contains $\theta$, but its distribution $\chi^2_{2n}$ does not depend on $\theta$.

**Step 3: invert.** Take the equal-tailed quantiles, cutting $\alpha/2$ from each tail:

$$\mathsf{P}\left(\chi^2_{2n,\alpha/2} \leq \frac{2n\overline{\mathbf{X}}}{\theta} \leq \chi^2_{2n,1-\alpha/2}\right) = 1 - \alpha.$$

Solve each inequality for $\theta$. Everything is positive, so taking reciprocals reverses the inequalities:

- $\dfrac{2n\overline{\mathbf{X}}}{\theta} \le \chi^2_{2n,1-\alpha/2} \iff \theta \ge \dfrac{2n\overline{\mathbf{X}}}{\chi^2_{2n,1-\alpha/2}}$
- $\dfrac{2n\overline{\mathbf{X}}}{\theta} \ge \chi^2_{2n,\alpha/2} \iff \theta \le \dfrac{2n\overline{\mathbf{X}}}{\chi^2_{2n,\alpha/2}}$

$$\theta \in \left[\frac{2n\overline{\mathbf{X}}}{\chi^2_{2n,1-\alpha/2}}, \frac{2n\overline{\mathbf{X}}}{\chi^2_{2n,\alpha/2}}\right]$$

Note that the **upper** quantile gives the **lower** endpoint.

**Properties.**

- **Exact.** No approximation was made anywhere, so the coverage is exactly $1-\alpha$ for every $n$, even $n=1$. The simulation confirms it: all values are within Monte Carlo noise of 0.90/0.95/0.99.
- **Length:** $L_1 = 2n\overline{\mathbf{X}}\left(\dfrac{1}{\chi^2_{2n,\alpha/2}} - \dfrac{1}{\chi^2_{2n,1-\alpha/2}}\right)$. It is random through $\overline{\mathbf{X}}$, and since $\mathsf{E}\overline{\mathbf{X}} = \theta$:

  $$\mathsf{E}L_1 = 2n\theta\left(\frac{1}{\chi^2_{2n,\alpha/2}} - \frac{1}{\chi^2_{2n,1-\alpha/2}}\right).$$

  Check: at $n=5$, $\alpha=0.05$, this formula gives 2.851, and the simulation gave 2.852.
- **Not symmetric around $\overline{\mathbf{X}}$.** The upper part is longer, because $\overline{\mathbf{X}}$ is right-skewed and the interval follows its true shape.
- **Large-$n$ behaviour.** $\chi^2_{2n}$ has mean $2n$ and variance $4n$, so by the CLT $\chi^2_{2n,p} \approx 2n + z_p \cdot 2\sqrt{n}$. Substituting,

  $$L_1 \approx 2n\overline{\mathbf{X}}\left(\frac{1}{2n-2z\sqrt{n}} - \frac{1}{2n+2z\sqrt{n}}\right) = \overline{\mathbf{X}} \cdot \frac{2z/\sqrt{n}}{1-z^2/n} \approx \frac{2z\overline{\mathbf{X}}}{\sqrt{n}},$$

  i.e. the same as the normal intervals. At $n=1000$ all methods give about 0.137.

**Code** (`exp_intervals`):

```
chi_lo, chi_hi = stats.chi2.ppf([alpha/2, beta], df=2*n)   # chi^2_{2n, α/2}, chi^2_{2n, 1-α/2}
lo1, hi1 = 2*n*xbar/chi_hi, 2*n*xbar/chi_lo
```

## 1.3 Method 2 — normal approximation with the known (true) variance

**Derivation.** By the CLT with $\mu = \theta$ and $\sigma = \theta$:

$$Z = \frac{\sqrt{n}(\overline{\mathbf{X}} - \theta)}{\theta} \approx \mathcal{N}(0,1) \implies \mathsf{P}(|Z| \le z_\beta) \approx 2\beta - 1 = 1 - \alpha.$$

Unpack $|Z| \le z_\beta$: multiply by $\theta/\sqrt{n} > 0$ to get $\left|\overline{\mathbf{X}} - \theta\right| \le z_\beta\theta/\sqrt{n}$. So

$$\theta \in \left[\overline{\mathbf{X}} - z_\beta\frac{\theta}{\sqrt{n}}, \overline{\mathbf{X}} + z_\beta\frac{\theta}{\sqrt{n}}\right]$$

**The problem with it.** The endpoints contain the **unknown $\theta$**, so this is not a statistic, and you cannot compute this interval from data. We used the pivot but skipped step 3 of the recipe (we did not solve for $\theta$). In the simulation we plug in the true $\theta$ only to have a **benchmark**: "what if we magically knew the variance".

**Length:** $L_2 = 2z_\beta\theta/\sqrt{n}$. It is **deterministic** (not random) and the shortest of the four. For $n = 5$, $\alpha = 0.05$: $2 \cdot 1.96 \cdot 1.1/\sqrt{5} = 1.928$. ✓

**Exact coverage of method 2 (bonus derivation in the notebook).** We do not need simulation here. $n\overline{\mathbf{X}}/\theta = \sum X_i/\theta = \sum E_i$ is a sum of $n$ independent $\mathscr{E}(1)$ variables, so it is $G \sim \Gamma(n, 1)$. Then

$$|Z| \le z_\beta \iff \left|\frac{n\overline{\mathbf{X}}}{\theta} - n\right| \le z_\beta\sqrt{n} \iff n - z_\beta\sqrt{n} \le G \le n + z_\beta\sqrt{n},$$

(multiply $\left|\overline{\mathbf{X}}/\theta - 1\right| \le z_\beta/\sqrt{n}$ by $n$), so

$$\mathsf{P}(\text{cover}) = F_{\Gamma(n,1)}(n + z_\beta\sqrt{n}) - F_{\Gamma(n,1)}\big(\max(n - z_\beta\sqrt{n}, 0)\big).$$

In the code this is `exact_coverage_23`. It is the "exact (2, 3)" column, and it agrees with the simulated values to within about 0.002.

**Why the error changes sign with $\alpha$.** $\Gamma(5, 1)$ is right-skewed: its left tail is short (it cannot go below 0) and its right tail is long. The normal approximation puts $\alpha/2$ in each tail. The real tail probabilities for $n = 5$ are:

| $\alpha$ | real left tail | real right tail | total miss | normal says | effect |
|---|---|---|---|---|---|
| 0.10 | 0.011 | 0.067 | 0.078 | 0.10 | over-cover (0.922) |
| 0.05 | 0.0004 | 0.043 | 0.044 | 0.05 | over-cover (0.956) |
| 0.01 | 0.0000 | 0.018 | 0.018 | 0.01 | **under**-cover (0.982) |

The left tail is always too small (it helps coverage) and the right tail is always too big (it hurts coverage). Which effect wins depends on how far out in the tail we cut, i.e. on $\alpha$.

## 1.4 Method 3 — solve the inequality for $\theta$

Start from the same event as in method 2 and do step 3 properly. Since $\theta > 0$:

$$\left|\theta - \overline{\mathbf{X}}\right| \le z_\beta\frac{\theta}{\sqrt{n}} \iff -z_\beta\frac{\theta}{\sqrt{n}} \le \theta - \overline{\mathbf{X}} \le z_\beta\frac{\theta}{\sqrt{n}}.$$

Split it into two inequalities and collect the $\theta$ terms. Write $c = z_\beta/\sqrt{n}$ for short.

- Left part: $-c\theta \le \theta - \overline{\mathbf{X}} \iff \overline{\mathbf{X}} \le \theta(1 + c) \iff \theta \ge \dfrac{\overline{\mathbf{X}}}{1 + c}$. Dividing by $1 + c > 0$ is always allowed.
- Right part: $\theta - \overline{\mathbf{X}} \le c\theta \iff \theta(1 - c) \le \overline{\mathbf{X}}$.
  - If $c < 1$ (i.e. $z_\beta < \sqrt{n}$): $1 - c > 0$, so $\theta \le \dfrac{\overline{\mathbf{X}}}{1 - c}$.
  - If $c \ge 1$: the left side $\theta(1 - c) \le 0 < \overline{\mathbf{X}}$, so the inequality holds **for every** $\theta > 0$ and there is no upper bound.

$$\theta \in \left[\frac{\overline{\mathbf{X}}}{1 + z_\beta/\sqrt{n}}, \frac{\overline{\mathbf{X}}}{1 - z_\beta/\sqrt{n}}\right] \quad (z_\beta < \sqrt{n}), \qquad \left[\frac{\overline{\mathbf{X}}}{1 + z_\beta/\sqrt{n}}, +\infty\right) \quad (z_\beta \ge \sqrt{n})$$

Example of the infinite case: $n = 5$, $\alpha = 0.01$, where $z_{0.995} = 2.576 > \sqrt{5} = 2.236$. That is why the table shows `inf`. In code: `hi3 = xbar/(1 - z/np.sqrt(n)) if z < np.sqrt(n) else np.full(m, np.inf)`.

**Coverage is identical to method 2.** We only rewrote the event $\left\{\left|\theta - \overline{\mathbf{X}}\right| \le z_\beta\theta/\sqrt{n}\right\}$ with equivalent ($\iff$) steps. On every single sample, "$\theta \in$ interval 2" and "$\theta \in$ interval 3" are the same event. That is why the columns "2" and "3" in the coverage table are **exactly equal**, not just

close: the same samples are used for all methods. The difference is that interval 3 is computable from data.

**Length.**

$$L_3 = \overline{\mathbf{X}}\left(\frac{1}{1-c} - \frac{1}{1+c}\right) = \overline{\mathbf{X}} \cdot \frac{(1+c)-(1-c)}{(1-c)(1+c)} = \overline{\mathbf{X}}\,\frac{2c}{1-c^2} = \overline{\mathbf{X}}\,\frac{2z_\beta/\sqrt{n}}{1-z_\beta^2/n}.$$

Taking the expectation ($\mathsf{E}\overline{\mathbf{X}} = \theta$):

$$\mathsf{E}L_3 = \frac{2z_\beta\theta/\sqrt{n}}{1-z_\beta^2/n} = \frac{L_2}{1-z_\beta^2/n} > L_2.$$

The factor $\frac{1}{1-z_\beta^2/n}$ blows up as $z_\beta^2 \to n$. For $n=5$, $\alpha=0.05$: $z^2/n = 0.768$, so the factor is 4.3 and $\mathsf{E}L_3 = 8.32$, against $L_2 = 1.93$. For large $n$ the factor tends to 1.

## 1.5 Method 4 — Student $t$ with the sample standard deviation

**Idea.** Replace the unknown $s = \theta$ in the denominator of $Z$ by its estimate $S$:

$$T = \frac{\sqrt{n}(\overline{\mathbf{X}} - \theta)}{S} \approx t_{n-1} \implies \mathsf{P}(|T| \le t_{n-1,\beta}) \approx 1-\alpha.$$

Since $S > 0$: $|T| \le t \iff \left|\overline{\mathbf{X}} - \theta\right| \le tS/\sqrt{n}$, so

$$\boxed{\theta \in \left[\overline{\mathbf{X}} - t_{n-1,\beta}\frac{S}{\sqrt{n}},\ \overline{\mathbf{X}} + t_{n-1,\beta}\frac{S}{\sqrt{n}}\right]}$$

**Where $t_{n-1}$ comes from (only exact for normal data).** If $X_i \sim \mathcal{N}(\mu, \sigma^2)$, then:

1. $Z = \sqrt{n}(\overline{\mathbf{X}} - \mu)/\sigma \sim \mathcal{N}(0,1)$ exactly.
2. $V = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. One degree of freedom is "used up" by estimating $\mu$ with $\overline{\mathbf{X}}$; see §3.4.
3. $\overline{\mathbf{X}}$ and $S^2$ are **independent**. This is a special property of the normal distribution (Cochran's theorem).

Then

$$\frac{Z}{\sqrt{V/(n-1)}} = \frac{\sqrt{n}(\overline{\mathbf{X}} - \mu)/\sigma}{S/\sigma} = \frac{\sqrt{n}(\overline{\mathbf{X}} - \mu)}{S} \sim t_{n-1}$$

(the unknown $\sigma$ cancels).

**Why it is only approximate for exponential data.** Points 1–3 all fail: $\overline{\mathbf{X}}$ is not normal (it is Gamma), $S^2$ is not scaled $\chi^2$, and most importantly **$\overline{\mathbf{X}}$ and $S$ are strongly dependent**. For $\mathscr{E}(\lambda)$ with $n=5$, $\operatorname{corr}(\overline{\mathbf{X}}, S) \approx 0.77$. For large $n$ it still works: $S \to \sigma$ (LLN), so by Slutsky $T \to \mathcal{N}(0,1)$, and $t_{n-1,\beta} \to z_\beta$.

**Why it under-covers (the most important observation in Problem 1).** Because of the positive correlation:

- A sample with an unusually **small** $\overline{\mathbf{X}}$ (the interval is shifted left, below $\theta$) usually also has a **small** $S$, so the interval is **narrow** and cannot reach $\theta$. It misses.
- A sample with a **large** $\overline{\mathbf{X}}$ has a large $S$ and a wide interval, so it still often covers.

So the misses on the left are not compensated. The coverage for $n=5$ is 0.83/0.88/0.95 instead of 0.90/0.95/0.99, and only for $n \gtrsim 200$ is it within about 0.005 of nominal.

**Length:** $L_4 = 2t_{n-1,\beta}S/\sqrt{n}$, which is random.

## 1.6 The simulation (part b), step by step

```
x = np.random.exponential(scale=theta, size=(m, n))   # m samples of size n, one per row
xbar = x.mean(axis=1)                                  # m sample means
s    = x.std(axis=1, ddof=1)                           # m sample std's with 1/(n-1)
...
coverage = np.mean((lo <= theta) & (theta <= hi))      # fraction of the m intervals hitting θ  = p̂
length   = np.mean(hi - lo)                            # estimate of E[L]
```

- `ddof=1` means "delta degrees of freedom": divide by $n - \texttt{ddof} = n - 1$. This is the unbiased version from Problem 3.
- All four methods use **the same** $m \times n$ matrix, so they are compared on identical data. This is why columns 2 and 3 coincide exactly.
- Grid: $n \in \{5, ..., 1000\}$, $\alpha \in \{0.1, 0.05, 0.01\}$, $m = 10\,000$.
- Effect of $m$: fix $n = 10$, $\alpha = 0.05$; for each $m \in \{10^2, ..., 10^5\}$ repeat the whole experiment 20 times and look at the spread (std) of $\hat{p}$. Compare it with the theoretical SE $\sqrt{p(1-p)/m}$: observed $0.019, 0.008, 0.002, 0.0005$ against theory $0.022, 0.007, 0.002, 0.0007$. ✓

## 1.7 Results and their explanation (parts c, d)

**Effect of $n$ on coverage.**

| Method | Behaviour | Theoretical reason |
|---|---|---|
| 1 | $= 1 - \alpha$ for all $n$ | exact pivot |
| $2 = 3$ | slightly off for small $n$, $\to 1 - \alpha$ | CLT error from the skewness of $\Gamma(n, 1)$, §1.3 |
| 4 | strongly **below** $1 - \alpha$ for small and moderate $n$ | dependence of $\overline{\mathbf{X}}$ and $S$, §1.5 |

**Effect of $n$ on length.** Everything is proportional to $1/\sqrt{n}$ (§0.2): from $n = 100$ to $n = 1000$ the length shrinks by about $\sqrt{10} \approx 3.16$. For small $n$ the order is $L_2 < L_4 < L_1 < L_3$ (at $n = 5$, $\alpha = 0.05$: $1.93 < 2.36 < 2.85 < 8.33$). For large $n$ all lengths coincide.

**Effect of $\alpha$.** A smaller $\alpha$ gives a larger quantile, so the interval is longer. This is the trade-off: more confidence costs precision.

**Effect of $m$.** None on the true coverage or the expected length. It only reduces the Monte Carlo noise (§0.8).

**Comparison table (part d).**

| Method | Pros | Cons | When to use |
|---|---|---|---|
| 1 | exact for every $n$; never degenerate; follows the skewness | needs the exact distribution, so only works for the exponential model | **best** whenever the data are known to be exponential |
| 2 | shortest, deterministic | **not computable** (needs $\theta$) | only a theoretical benchmark |
| 3 | computable; coverage = method 2's | huge or infinite for small $n$; needs the model $s^2 = \theta^2$ | moderate $n$, variance a known function of the mean |
| 4 | universal (any distribution with finite variance) | under-covers for skewed data until $n$ is in the hundreds | unknown distribution, large $n$ |

# 2. Problem 2 — Poisson distribution

## 2.1 Setup and moments

$X_1, ..., X_n \sim \mathscr{P}(\theta)$, with $\theta = 1.1$ (and also 5 and 20 for comparison).

**Derivation of $\mathsf{E}X = \theta$, $\operatorname{Var} X = \theta$.**

$$\mathsf{E}X = \sum_{k=0}^{\infty} k\frac{e^{-\theta}\theta^k}{k!} = \theta e^{-\theta}\sum_{k=1}^{\infty}\frac{\theta^{k-1}}{(k-1)!} = \theta e^{-\theta}e^{\theta} = \theta.$$

The same trick with $k(k-1)$:

$$\mathsf{E}X(X-1) = \sum_{k=2}^{\infty} k(k-1)\frac{e^{-\theta}\theta^k}{k!} = \theta^2 e^{-\theta}\sum_{k=2}^{\infty}\frac{\theta^{k-2}}{(k-2)!} = \theta^2.$$

So $\mathsf{E}X^2 = \theta^2 + \theta$ and $\operatorname{Var} X = \theta^2 + \theta - \theta^2 = \theta$.

So here $s^2 = \theta$, $s = \sqrt{\theta}$. The variance again depends on the unknown mean, but now as $\sqrt{\theta}$ instead of $\theta$.

**Why there is no "method 1" for Poisson.** The task says to repeat parts (2)–(4) only. The Poisson distribution is discrete, so there is no continuous pivot with a $\theta$-free distribution like $\chi^2_{2n}$. (An "exact" conservative interval does exist, the Garwood interval, built from $\chi^2$ quantiles via the Poisson–Gamma relation, but it is not part of this task.)

**Useful fact.** $T = \sum X_i \sim \mathscr{P}(n\theta)$, because a sum of independent Poissons is Poisson (MGF: $e^{\theta(e^t-1)}$ to the power $n$). $T$ takes only **integer** values. This explains the "zigzag" coverage.

## 2.2 Method 2 — normal approximation, known variance

By the CLT with $\mu = \theta$ and $\sigma = \sqrt{\theta}$:

$$Z = \frac{\sqrt{n}(\overline{\mathbf{X}} - \theta)}{\sqrt{\theta}} \approx \mathcal{N}(0,1) \implies \mathsf{P}\left(\left|\overline{\mathbf{X}} - \theta\right| \le z_\beta\sqrt{\theta/n}\right) \approx 1 - \alpha$$

(multiply $|Z| \le z_\beta$ by $\sqrt{\theta}/\sqrt{n}$). Hence

$$\theta \in \left[\overline{\mathbf{X}} - z_\beta\sqrt{\theta/n},\ \overline{\mathbf{X}} + z_\beta\sqrt{\theta/n}\right]$$

As before, this depends on the unknown $\theta$, so it is a benchmark only. Its length $L_2 = 2z_\beta\sqrt{\theta/n}$ is deterministic. For $n = 5$, $\alpha = 0.05$: $2 \cdot 1.96\sqrt{1.1/5} = 1.839$. ✓

**Exact coverage (verification).** Multiply the event by $n$: $|T - n\theta| \le z_\beta\sqrt{n\theta}$, with $T \sim \mathscr{P}(n\theta)$. So the coverage depends only on $n\theta$ and equals a **sum of Poisson probabilities over the integers** inside $[n\theta - z\sqrt{n\theta},\ n\theta + z\sqrt{n\theta}]$. For $\theta = 1.1$:

| $\alpha$ | $n$ | allowed $T$ | exact coverage | simulated |
|---|---|---|---|---|
| 0.10 | 5 | $2, ..., 9$ | 0.9197 | 0.9177 |
| 0.10 | 20 | $15, ..., 29$ | 0.8921 | 0.8950 |
| 0.05 | 5 | $1, ..., 10$ | 0.9707 | 0.9701 |
| 0.05 | 10 | $5, ..., 17$ | 0.9527 | 0.9569 |
| 0.01 | 5 | $0, ..., 11$ | 0.9890 | 0.9882 |

Because $T$ is an integer, as $n$ changes the interval endpoints jump over integers. The coverage therefore jumps up and down: it is **not monotone in $n$** (for example, $\alpha = 0.1$ gives 0.9197 at $n = 5$ but 0.8921 at $n = 20$). This is the discreteness effect mentioned in the conclusions.

### 2.3 Method 3 — solve the inequality (a quadratic now)

Start from $\left|\theta - \overline{\mathbf{X}}\right| \le z_\beta \sqrt{\theta/n}$. Both sides are $\ge 0$, so **squaring is an equivalence**:

$$(\theta - \overline{\mathbf{X}})^2 \le z_\beta^2 \frac{\theta}{n}.$$

Expand and collect the powers of $\theta$:

$$\theta^2 - 2\overline{\mathbf{X}}\theta + \overline{\mathbf{X}}^2 - \frac{z_\beta^2}{n}\theta \le 0 \iff \theta^2 - \left(2\overline{\mathbf{X}} + \frac{z_\beta^2}{n}\right)\theta + \overline{\mathbf{X}}^2 \le 0.$$

This is a quadratic in $\theta$ with positive leading coefficient (an upward parabola), so it is $\le 0$ exactly **between its roots**.

**Discriminant.** With $b = 2\overline{\mathbf{X}} + z^2/n$ and $c = \overline{\mathbf{X}}^2$:

$$D = b^2 - 4c = 4\overline{\mathbf{X}}^2 + \frac{4\overline{\mathbf{X}}z^2}{n} + \frac{z^4}{n^2} - 4\overline{\mathbf{X}}^2 = \frac{4\overline{\mathbf{X}}z^2}{n} + \frac{z^4}{n^2} \ge 0,$$

so the roots are always real.

**Roots:**

$$\theta_{1,2} = \frac{b \pm \sqrt{D}}{2} = \overline{\mathbf{X}} + \frac{z_\beta^2}{2n} \pm \frac{1}{2}\sqrt{\frac{4\overline{\mathbf{X}}z_\beta^2}{n} + \frac{z_\beta^4}{n^2}} = \overline{\mathbf{X}} + \frac{z_\beta^2}{2n} \pm z_\beta\sqrt{\frac{\overline{\mathbf{X}}}{n} + \frac{z_\beta^2}{4n^2}}.$$

$$\theta \in [\theta_1, \theta_2], \qquad \theta_{1,2} = \overline{\mathbf{X}} + \frac{z_\beta^2}{2n} \pm z_\beta\sqrt{\frac{\overline{\mathbf{X}}}{n} + \frac{z_\beta^2}{4n^2}}$$

**Checks.**

- **Both roots are $\ge 0$.** By Vieta's formulas, $\theta_1\theta_2 = \overline{\mathbf{X}}^2 \ge 0$ and $\theta_1 + \theta_2 = b > 0$. So the interval lies in the valid parameter range, which is needed because $\sqrt{\theta}$ only makes sense for $\theta \ge 0$.
- **The centre is shifted** from $\overline{\mathbf{X}}$ to $\overline{\mathbf{X}} + z^2/(2n)$, to the right. This matches the right skewness of the Poisson distribution for small $\theta$.
- **Never degenerate.** If $\overline{\mathbf{X}} = 0$ (all observations are 0), the roots are 0 and $z^2/n$, so the interval is $[0, z_\beta^2/n]$, which is still a sensible interval. Compare with method 4 below.
- **Length:** $L_3 = 2z_\beta\sqrt{\overline{\mathbf{X}}/n + z_\beta^2/(4n^2)}$, random. As $n \to \infty$ the $z^2/(4n^2)$ term vanishes faster than $\overline{\mathbf{X}}/n$, so $L_3 \approx 2z\sqrt{\theta/n} = L_2$.
- **Same coverage as method 2,** for exactly the same reason as in Problem 1: squaring was an equivalence, so on each sample the events coincide. The tables show identical columns 2 and 3.

(This construction is the Poisson analogue of the **Wilson score interval** for a binomial proportion.)

**Code:**

```
centre3 = xbar + z**2 / (2*n)
h3 = z * np.sqrt(xbar/n + z**2 / (4*n**2))
lo3, hi3 = centre3 - h3, centre3 + h3
```

## 2.4 Method 4 — Student $t$

Exactly as in Problem 1:

$$\theta \in \left[\overline{\mathbf{X}} - t_{n-1,\beta}\frac{S}{\sqrt{n}}, \overline{\mathbf{X}} + t_{n-1,\beta}\frac{S}{\sqrt{n}}\right]$$

This does not use the Poisson model at all. Its specific problems for Poisson data:

- **Degeneration.** If all $X_i = 0$, then $\overline{\mathbf{X}} = S = 0$ and the interval is the single point $\{0\}$, which never contains $\theta > 0$. The probability is $\mathsf{P}(T = 0) = e^{-n\theta} = e^{-5.5} \approx 0.004$ for $n = 5$, $\theta = 1.1$: small, but it is a guaranteed miss.
- With small $n\theta$, the sample contains only a few distinct small integers. Then $S$ is a poor estimate of $\sqrt{\theta}$, and it is again positively correlated with $\overline{\mathbf{X}}$ (for Poisson, mean = variance). The result is under-coverage for small $n$, for example 0.905 instead of 0.95 at $n = 5$.

## 2.5 Results and explanation

**Coverage** ($\alpha = 0.05$, $\theta = 1.1$):

| $n$ | 5 | 10 | 20 | 50 | 100 | 1000 |
|---|---|---|---|---|---|---|
| $2 = 3$ | 0.970 | 0.957 | 0.961 | 0.952 | 0.948 | 0.955 |
| 4 | 0.905 | 0.945 | 0.946 | 0.948 | 0.946 | 0.953 |

**Length** ($\alpha = 0.05$): at $n = 5$, $L_2 = 1.84 < L_3 = 1.96 < L_4 = 2.42$. At $n = 1000$ all are 0.130.

**Explanations.**

- ***n*:** the coverage tends to $1 - \alpha$ and the length shrinks like $1/\sqrt{n}$, as in Problem 1. The coverage is not smooth in $n$ because of discreteness (§2.2).
- ***m*:** only one run per $m$ here, at $n = 50$. At $m = 100$ the estimates (0.93/0.94) are off by about 1 SE ($\approx 0.022$), which is pure noise. For $m \geq 10^4$ they are stable.
- ***θ* (unlike Problem 1, it matters here).** The Poisson family is **not** a scale family, so the coverage depends on $\theta$; for methods 2 and 3 it depends only on $n\theta$ (§2.2). For $\theta = 5$ or 20, $n\theta$ is large even for $n = 5$. The Poisson distribution is then nearly symmetric and normal, so all methods are within about 0.01 of nominal right away.
- ***α*:** the deviation from nominal is not monotone in $\alpha$. Whether the endpoints happen to land just before or just after an integer value of $T$ decides it.

**Best method.** Method 3 if the data are known to be Poisson, especially for small $n$ or small $\theta$: it is computable, has good coverage, and never degenerates. Method 4 if the model is unknown and $n$ is large. For large $n$ all methods are practically equivalent. Method 2 is a benchmark only.

# 3. Problem 3 — Unbiasedness of variance estimators

## 3.1 The two estimators

$$\sigma_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2, \qquad \sigma_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Both use the same sum of squares $SS = \sum(X_i - \overline{X})^2$ and differ only in the divisor. On any single sample, $\sigma_{n-1}^2 = \frac{n}{n-1}\sigma_n^2$, so $\sigma_{n-1}^2$ is always slightly larger.

In NumPy: `np.var(x, ddof=0)` gives $\sigma_n^2$, and `np.var(x, ddof=1)` gives $\sigma_{n-1}^2$.

**Parameters:** $\mu = \texttt{ID} = 11$, $\sigma^2 = (11 \bmod 5) + 2 = 1 + 2 = 3$. In code, `np.random.normal(loc=MU, scale=SIGMA)`: note that `scale` is the **standard deviation** $\sqrt{3}$, not the variance.

## 3.2 Parts (a)–(c): what the code does

- **(a)** Generates 1000 values from $\mathcal{N}(11, 3)$. Sample mean 11.036, variance 3.041, both close to the true values, so the generator is set up correctly.
- **(b)** Computes both estimators on **one** sample (the first $n$ values of the dataset):

| $n$ | 10 | 50 | 100 | 1000 |
|---|---|---|---|---|
| $\sigma_n^2$ | 2.398 | 3.884 | 3.500 | 3.041 |
| $\sigma_{n-1}^2$ | 2.664 | 3.963 | 3.536 | 3.044 |

  Single estimates scatter around 3. This is **variance** of the estimator (random error), not bias. For normal data, $\mathrm{SD}(\sigma_{n-1}^2) = \sigma^2\sqrt{2/(n-1)}$ (see §3.6). At $n = 50$ that is $3\sqrt{2/49} \approx 0.61$, so 3.96 is only about 1.6 SD away, which is completely normal. At $n = 10$ the ratio between the two estimators is $10/9 \approx 1.11$; at $n = 1000$ it is 1.001.

- **(c)** Estimates $\mathsf{E}(\cdot)$ by averaging each estimator over $M = 10\,000$ independent datasets of size $n$ (an $M \times n$ matrix, `var(axis=1)`). Then bias = average − 3.

| $n$ | sim. Bias$(\sigma_n^2)$ | theory $-3/n$ | sim. Bias$(\sigma_{n-1}^2)$ | theory |
|---|---|---|---|---|
| 10 | −0.309 | −0.300 | −0.010 | 0 |
| 50 | −0.050 | −0.060 | +0.010 | 0 |
| 100 | −0.027 | −0.030 | +0.003 | 0 |
| 1000 | −0.002 | −0.003 | +0.001 | 0 |

  **Are the differences from theory just noise?** The Monte Carlo SE of an average over $M$ datasets is $\mathrm{SD}(\text{estimator})/\sqrt{M}$. For $\sigma_{n-1}^2$ at $n = 10$ this is $3\sqrt{2/9}/100 \approx 0.014$, so −0.010 is well within 1 SE. At $n = 50$: $\approx 0.006$, so the deviations of about 0.01 are about 1.6 SE. Yes, they are noise, and they have no fixed sign.

## 3.3 Part (e): derivation of $\mathsf{E}(\sigma_n^2)$ and $\mathsf{E}(\sigma_{n-1}^2)$

Assumptions: $X_i$ i.i.d., $\mathsf{E}X_i = \mu$, $\mathrm{Var}\,X_i = \sigma^2$. **Normality is not needed**, so the result holds for any distribution with finite variance.

**Step 1 — rewrite $SS$ through deviations from the true mean $\mu$.** Insert $\pm\mu$:

$$X_i - \overline{X} = (X_i - \mu) - (\overline{X} - \mu).$$

Square, using $(a-b)^2 = a^2 - 2ab + b^2$, and sum over $i$. Note that $(\overline{X} - \mu)$ does not depend on $i$, so it comes out of the sum:

$$SS = \sum_{i=1}^{n} (X_i - \mu)^2 - 2(\overline{X} - \mu) \sum_{i=1}^{n} (X_i - \mu) + n(\overline{X} - \mu)^2.$$

The middle sum is $\sum (X_i - \mu) = \sum X_i - n\mu = n\overline{X} - n\mu = n(\overline{X} - \mu)$. So the middle term is $-2n(\overline{X} - \mu)^2$, and combining it with the last term:

$$SS = \sum_{i=1}^{n} (X_i - \mu)^2 - n(\overline{X} - \mu)^2.$$

**Step 2 — take expectations term by term** (linearity):

- $\mathsf{E}(X_i - \mu)^2 = \operatorname{Var} X_i = \sigma^2$ by the definition of variance, because $\mathsf{E}X_i = \mu$. So $\mathsf{E} \sum (X_i - \mu)^2 = n\sigma^2$.
- $\mathsf{E}(\overline{X} - \mu)^2 = \operatorname{Var} \overline{X}$, because $\mathsf{E}\overline{X} = \mu$. By §0.2 this is $\sigma^2/n$ (here independence is used).

$$\mathsf{E}(SS) = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2.$$

**Step 3 — divide:**

$$\boxed{\mathsf{E}(\sigma_n^2) = \frac{\mathsf{E}(SS)}{n} = \frac{n-1}{n}\sigma^2, \qquad \mathsf{E}(\sigma_{n-1}^2) = \frac{\mathsf{E}(SS)}{n-1} = \sigma^2}$$

**Alternative derivation (good to know if asked "do it another way").** Use $\mathsf{E}Y^2 = \operatorname{Var} Y + (\mathsf{E}Y)^2$:

- $\mathsf{E}X_i^2 = \sigma^2 + \mu^2$
- $\mathsf{E}\overline{X}^2 = \sigma^2/n + \mu^2$

Expand directly:

$$SS = \sum X_i^2 - 2\overline{X} \sum X_i + n\overline{X}^2 = \sum X_i^2 - 2n\overline{X}^2 + n\overline{X}^2 = \sum X_i^2 - n\overline{X}^2.$$

$$\mathsf{E}(SS) = n(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right) = n\sigma^2 - \sigma^2 = (n-1)\sigma^2. \checkmark$$

## 3.4 Part (f): which estimator is unbiased

$$\operatorname{Bias}(\sigma_n^2) = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{n} \neq 0, \qquad \operatorname{Bias}(\sigma_{n-1}^2) = \sigma^2 - \sigma^2 = 0.$$

- $\sigma_{n-1}^2$ is **unbiased** for every $n \geq 2$ and every distribution with finite variance.
- $\sigma_n^2$ is **biased downward**: it underestimates by $\sigma^2/n$ on average. Since $-\sigma^2/n \to 0$, it is **asymptotically unbiased**. For $\sigma^2 = 3$: $-0.3, -0.06, -0.03, -0.003$ for $n = 10, 50, 100, 1000$.

**Why it is biased (intuition, likely to be asked).**

1. **$\overline{X}$ minimises the sum of squares.** $g(a) = \sum (X_i - a)^2$ has $g'(a) = -2\sum (X_i - a) = 0 \iff a = \overline{X}$. So for every sample, $\sum (X_i - \overline{X})^2 \leq \sum (X_i - \mu)^2$: measuring spread around the fitted centre $\overline{X}$ always looks smaller than around the true centre $\mu$. Step 1 shows the gap is exactly $n(\overline{X} - \mu)^2$, whose expectation is $n \cdot \sigma^2/n = \sigma^2$, i.e. we lose exactly one $\sigma^2$.
2. **Degrees of freedom.** The $n$ deviations $X_i - \overline{X}$ are not free: they always sum to 0, so knowing $n-1$ of them determines the last one. Only $n-1$ independent pieces of information about the

spread remain. One was "spent" on estimating $\mu$. Dividing by the number of free deviations, $n-1$, gives the right average. This correction is called **Bessel's correction**.

3. **If $\mu$ were known,** then $\frac{1}{n}\sum(X_i-\mu)^2$ would be unbiased (from Step 2: its expectation is $\sigma^2$). The $n-1$ is needed only because $\mu$ is replaced by $\overline{X}$.

### 3.5 Parts (d) and (g): conclusions in one paragraph

Single-sample estimates (b) scatter randomly around $\sigma^2 = 3$, with a scatter that decreases with $n$. Averaged over 10 000 datasets (c), $\sigma_n^2$ is systematically below 3 by almost exactly $3/n$, and $\sigma_{n-1}^2$ is at 3 up to Monte Carlo noise. The simulation therefore confirms the theory $\mathrm{Bias}(\sigma_n^2) = -\sigma^2/n$, $\mathrm{Bias}(\sigma_{n-1}^2) = 0$: on the plot, the simulated $\sigma_n^2$ curve lies on the dashed line $-\sigma^2/n$. For small $n$, use $\sigma_{n-1}^2$. For large $n$ the two are practically identical, because their ratio is $n/(n-1) \to 1$.

### 3.6 Extra (beyond the notebook, useful for defence questions): unbiased $\neq$ "better"

For **normal** data, $(n-1)\sigma_{n-1}^2/\sigma^2 \sim \chi_{n-1}^2$. A $\chi_k^2$ variable has mean $k$ and variance $2k$. Hence:

$$\mathrm{Var}(\sigma_{n-1}^2) = \frac{\sigma^4}{(n-1)^2}\cdot 2(n-1) = \frac{2\sigma^4}{n-1}, \qquad \mathrm{MSE}(\sigma_{n-1}^2) = \frac{2\sigma^4}{n-1}.$$

$$\mathrm{Var}(\sigma_n^2) = \left(\frac{n-1}{n}\right)^2 \frac{2\sigma^4}{n-1} = \frac{2(n-1)\sigma^4}{n^2}, \qquad \mathrm{MSE}(\sigma_n^2) = \frac{2(n-1)\sigma^4}{n^2} + \frac{\sigma^4}{n^2} = \frac{(2n-1)\sigma^4}{n^2}.$$

Since $(2n-1)(n-1) = 2n^2 - 3n + 1 < 2n^2$, we have $\mathrm{MSE}(\sigma_n^2) < \mathrm{MSE}(\sigma_{n-1}^2)$. The **biased** estimator has a **smaller** mean squared error: it trades a little bias for less variance. Unbiasedness is one desirable property, not the only one. (Also, $\sigma_n^2$ is the maximum-likelihood estimator for normal data.)

Also note: $S = \sqrt{\sigma_{n-1}^2}$ is **not** unbiased for $\sigma$. By Jensen's inequality, $\mathsf{E}S \le \sqrt{\mathsf{E}S^2} = \sigma$.

# 4. Likely defence questions

**What is a confidence interval? What does 95% mean?**
A random interval $[L, U]$ with $\mathsf{P}(L \le \theta \le U) = 0.95$. Over many repetitions, 95% of such intervals contain the fixed true $\theta$.

**Why $\beta = 1 - \alpha/2$?**
So that $\mathsf{P}(|Z| \le z_\beta) = 2\beta - 1 = 1 - \alpha$, i.e. $\alpha/2$ in each tail.

**Show that $2\lambda n\overline{\mathbf{X}} \sim \chi^2_{2n}$.**
CDF of $2\lambda X_i$ is $1 - e^{-y/2}$, which is $\chi^2_2$. MGF $(1-2t)^{-1}$; a product of $n$ of them is $(1-2t)^{-n}$, which is $\chi^2_{2n}$.

**Why is method 2 not a real confidence interval?**
Its endpoints contain the unknown $\theta$, so they cannot be computed from data.

**Why do methods 2 and 3 have exactly the same coverage?**
Method 3 is obtained from method 2′s event by equivalent transformations. On each sample the events coincide, and the same samples are used for both.

**Why can interval 3 be infinite in Problem 1?**
When $z_\beta \ge \sqrt{n}$, the coefficient $1 - z_\beta/\sqrt{n} \le 0$, so the upper inequality holds for every $\theta > 0$. Example: $n = 5$, $\alpha = 0.01$.

**Why does the Student-$t$ interval under-cover for exponential data?**
$t_{n-1}$ is exact only for normal data, where $\overline{\mathbf{X}}$ and $S$ are independent. For skewed data they are positively correlated, so low-$\overline{\mathbf{X}}$ samples also get narrow intervals and miss $\theta$.

**How do $n$ and $m$ affect the results?**
$n$ changes the true coverage (approximation quality) and the length ($\propto 1/\sqrt{n}$). $m$ changes neither. It only reduces the Monte Carlo error $\sqrt{p(1-p)/m}$.

**Why doesn't $\theta$ matter in Problem 1 but does in Problem 2?**
The exponential distribution is a scale family ($X = \theta E$), so all pivots are $\theta$-free. The Poisson distribution is not: its shape changes with $\theta$, and the coverage depends on $n\theta$.

**Derive the Poisson method-3 interval.**
Square $\left|\theta - \overline{\mathbf{X}}\right| \le z\sqrt{\theta/n}$, get the quadratic $\theta^2 - (2\overline{\mathbf{X}} + z^2/n)\theta + \overline{\mathbf{X}}^2 \le 0$, and take $\theta$ between its roots.

**What happens when all Poisson observations are 0?**
Method 3 gives $[0, z^2/n]$. Method 4 collapses to $\{0\}$ and always misses.

**Prove $\mathsf{E}\sum(X_i - \overline{X})^2 = (n-1)\sigma^2$.**
$SS = \sum(X_i - \mu)^2 - n(\overline{X} - \mu)^2$, and its expectation is $n\sigma^2 - n \cdot \sigma^2/n$.

**Why divide by $n - 1$?**
The deviations from $\overline{X}$ sum to 0, so there are $n - 1$ degrees of freedom. $\overline{X}$ is closer to the data than $\mu$, which loses exactly $\sigma^2$ on average.

**Did the proof use normality?**
No. It only used independence and finite variance.

**Is the unbiased estimator always better?**
No. For normal data, $\sigma^2_n$ has a smaller MSE (§3.6).

**What does `ddof=1` mean?**
Divide by $n - \texttt{ddof} = n - 1$.

**Why fix the seed?**
Reproducibility: rerunning the notebook gives identical random numbers and therefore identical results.